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DATA DESCRIPTOR

TCMEval-PA: a question-answering benchmark dataset for the prescription audit of Traditional Chinese Medicine

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Although large language models (LLMs) have witnessed rapid development in medical applications, their capacities to support rational medication use and guarantee prescription safety remain insufficiently investigated—especially in tasks such as prescription audit, which plays a critical role in safeguarding both. This paper presents TCMEval-PA, a benchmark dataset for assessing the capabilities of LLMs in prescription audit of Chinese herbal medicines. The dataset comprises 328 choice questions, including 297 single-choice and 31 multiple-choice. All questions were designed and compiled through rule extraction from official documents and reviewed by licensed TCM physicians. TCMEval-PA comprehensively encompasses the key dimensions of prescription safety, including normativity (e.g., dispensing, decoction requirements, and regulations for special medicines) and appropriateness (e.g., contraindicated combinations and excessive dosages). The present study employed TCMEval-PA to assess several prevalent Chinese and English LLMs. This dataset can be utilized for the evaluation of LLMs and other artificial intelligence (AI) systems in TCM prescription safety scenarios and promotes research in intelligent auditing and decision support.

Background & Summary

With their strong language comprehension¹, advanced text analysis capabilities², and extensive knowledge base, large language models (LLMs) have been increasingly adopted in the medical domain for medical question answering³, health consultation^{4,5}, clinical decision support⁶, and disease management⁷. Meanwhile, the emergence of vertical-domain models tailored to day dosage scenarios reflects the growing demand within the medical community for domain-specific AI technologies⁸. Despite the impressive capabilities that LLMs have demonstrated across various medical tasks⁹, their real-world deployment in clinical settings faces significant challenges, chief among them the issue of hallucination¹⁰. In the medical context, hallucinations may present as fabricated drug recommendations¹¹ or falsified clinical evidence. Such outputs not only compromise the reliability of LLM-generated content, but may also lead directly to misdiagnosis, inappropriate treatment, or harmful clinical decisions^{12,13}. Therefore, in scenarios where LLMs are used to provide medical advice, there is an urgent need for systematic evaluation of their reliability in ensuring rational medication use and prescription safety.

As a critical measure to ensure prescription quality and support rational medication use, prescription auditing entails a systematic evaluation of clinical prescriptions for legality, standardization, and appropriateness. Compared to those in Western pharmaceuticals, audits of Traditional Chinese Medicine (TCM) prescriptions are notably more intricate and challenging. While audits in Western medicine primarily address common prescription errors—such as inappropriate drug selection, incorrect dosage, or improper treatment duration¹⁴—Chinese herbal medicines prescriptions reflect a holistic approach, emphasizing individualized syndrome

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differentiation and synergistic effects among the herbal components, and prescription audits must additionally account for discipline-specific considerations. For instance, the classical TCM principle of the “Eighteen Incompatible Medicaments” (Shiba Fan)^{15,16} outlines specific contraindicated combinations that may induce or exacerbate toxicity and must therefore be avoided. The dosage of toxic Chinese herbal medicines must be strictly regulated, and prescriptions are also required to specify appropriate decoction and processing methods for specific herbs. Consequently, prescription audit of Chinese herbal medicines must address a broad array of elements, including but not limited to contraindicated combinations, dosage of individual Chinese herbal medicines, toxic Chinese herbal medicines, and specifications for preparation and administration. These distinctive requirements place heightened demands on LLMs, challenging their domain-specific knowledge accuracy and reasoning ability when applied to the prescription audit of Chinese herbal medicines.

As the public increasingly relies on LLMs for health guidance and medical advice, it is essential to systematically evaluate their ability to ensure rational medication use and prescription safety. Existing evaluations of TCM-oriented LLMs have largely focused on tasks such as medical examination question answering^{17,18}, syndrome differentiation reasoning^{19,20}, acupoint knowledge extraction and representation²¹, and prescription generation²². Recent work has started to consider medication safety in the TCM context²³; however, evaluations often remain at the level of isolated knowledge verification and lack a structured evaluation framework grounded in prescription audit standards. Given that prescription audits are directly linked to high-risk clinical issues, and that clinical settings often face staffing constraints, there is a growing need for LLM-based automated audit solutions. In this context, as a critical task closely tied to clinical medication compliance and risk prevention, the prescription auditing capabilities of LLMs urgently require a comprehensive benchmark that is aligned with authoritative standards.

To address this gap, we developed TCMEval-PA, a benchmark dataset comprising 328 choice questions, designed to evaluate the capabilities of LLMs in conducting the prescription audit of Chinese herbal medicines. TCMEval-PA comprehensively covers two fundamental dimensions essential to prescription safety: (1) Normativity, which examines whether prescriptions comply with national and industry standards in terms of structure, formatting, labeling, dosage, and the use of high-risk herbal materials; and (2) Appropriateness, which evaluates whether the use of Chinese herbal medicines is rational, safe, and consistent with TCM diagnostic principles and individual patient characteristics. All questions were developed based on authoritative sources, such as the *Technical Specification for Prescription Audit and Comment of Chinese Medicine Prescriptions*, and were independently annotated and cross-verified by licensed TCM physicians to ensure accuracy and consistency.

This study has two primary objectives:

1. To construct a high-quality dataset that comprehensively covers the full set of rules governing the prescription audit of Chinese herbal medicines.
2. To provide an evidence-based evaluation of the reliability of leading Chinese and English LLMs in supporting rational medication use, and to identify their specific limitations in this context.

This study is among the first to evaluate the performance of LLMs in auditing prescriptions of Chinese herbal medicines. Unlike traditional question–answering (QA)-style tasks that focus primarily on factual recall, TCMEval-PA is designed to evaluate a model’s ability to understand formalized rules, make compliance judgments, and apply standardized auditing criteria. Each question is explicitly mapped to rule clauses defined in national TCM prescription standards, with structured annotations that reflect the regulatory basis of the task. This design makes TCMEval-PA more aligned with the real-world demands of TCM prescription audit in terms of both task format and knowledge dimensions. These evaluations also help pinpoint areas where models fall short in rule compliance, thereby laying a solid foundation for the practical application of AIGC technologies in TCM-related intelligent auditing and pharmaceutical management.

Methods

Audit rule extraction. This dataset was designed in strict conformity with validated standards for the prescription audit of Chinese herbal medicines, thereby ensuring both regulatory compliance and clinical applicability of the questions. The *Technical Specification for Prescription Audit and Comment of Chinese Medicine Prescriptions* served as the primary reference, supplemented by national and industry publications such as the *Pharmacopoeia of the People’s Republic of China* (ChP) and the *Measures for the Administration of Toxic Drugs for Medical Use*. Drawing on these sources, we systematically extracted a total of 18 audit rules, each covering key safety dimensions relevant to the audit of prescriptions involving Chinese herbal medicines. Their emphasis led to the classification of these rules into two primary dimensions: prescription normativity (9 rules) and prescription appropriateness (9 rules). Detailed descriptions of the prescription audit rules are provided in Table 1 (with the original Chinese version available in Supplementary Table S1). Notably, the column “Prescription Audit Rule” in Table 1 presents a concise summary or title of each audit rule, which corresponds to the “Rule Summary” field in Table 4. These rules served as the foundation for dataset construction, question design, and annotation.

A glossary of standardized terms used in the audit rules is provided at the end of the manuscript.

Data collection and construction. Following the prescription audit rules outlined above, we systematically collected and constructed multiple-choice questions corresponding to each rule. The dataset construction workflow is illustrated in Fig. 1. The questions were primarily derived from authoritative training manuals and officially published examination books prepared for the Licensed Pharmacist Examination (Chinese Pharmacy) and the National Professional Technical Qualification Examination (Chinese Pharmacy)^{24–30}. These sources were selected to ensure authority and representativeness. All materials were obtained in print format and are not

Audit Dimension	Prescription Audit Rules	Rule Identifier ^a	Rule Description
Fundamental Concepts	Fundamental Concepts in Prescription Audit	C-001	The fundamental concepts of prescription audit encompass both the content and the institutional framework of auditing practices.
Normativity	Completeness of Prescription Documentation: Header, Main Body, and Footer Sections	N-001	Header: (1) General information includes the name of the medical institution, payment category, patient's name, sex, age, outpatient or inpatient medical record number, department or ward, bed number, and prescription date. (2) For prescriptions involving Chinese herbal medicines, the TCM diagnosis—including the disease name (optional if undetermined) and the syndrome pattern—must be clearly and completely documented, and must align with the patient's medical record. Main Body: Includes the names of prescribed Chinese herbal medicines, dosage of individual Chinese herbal medicines, number of doses, and administration instructions. Footer: (1) The prescribing physician must sign the prescription (either handwritten or electronic signature) or affix an official seal. (2) The total medication cost must be indicated. (3) Any modifications must be signed and dated by the prescriber.
	The prescription formulation should reflect the combinatorial hierarchy of “principal, associate, adjuvant, and envoy” (Jun–Chen–Zuo–Shi), in alignment with TCM compatibility principles.	N-002	The documentation of prescriptions containing Chinese herbal medicines should reflect the therapeutic hierarchy of <i>Jun–Chen–Zuo–Shi</i> (principal, associate, adjuvant, and envoy), in accordance with the foundational principles of TCM.
	Special dispensing and decoction requirements must be clearly annotated	N-003	Special instructions regarding dispensing and decoction should be indicated in the upper right corner of the corresponding medicine name and enclosed in parentheses.
	Use of Legally Approved Dosage Units	N-004	Dosages shall be written using legal units of measurement, expressed in Arabic numerals. In principle, grams (g) should be used as the standard unit, with the unit symbol “g” placed immediately after the numeric value.
	Prescription Dosage Regulations	N-005	Verify whether the prescription exceeds the standard 7-day supply, or the 3-day limit for emergency prescriptions. If the prescription duration exceeds the regulatory threshold, issue a clear alert and recommend that the prescriber modify the prescription. For cases involving chronic diseases, geriatric patients, or other special clinical scenarios, extended quantities may be permissible, provided that the prescribing physician has explicitly documented the justification within the prescription. For prescriptions issued to patients eligible for long-term prescriptions, confirm that the prescription duration complies with the <i>Long-Term Prescription Management Guidelines</i> . All such evaluations should be conducted as part of the prescription audit of Chinese herbal medicines.
	Prescription Validity Period Regulations	N-006	In principle, prescriptions are valid only on the day of issuance. Under special circumstances where deferred dispensing is required, the prescribing physician must explicitly indicate a validity period on the prescription form, which shall not exceed three days from the date of issuance. Prescriptions exceeding this validity period shall be deemed noncompliant and will be disqualified during the prescription audit process.
	Regulations on the Use of Special Medications	N-007	(1) Verify whether prescriptions involving narcotic drugs comply with the Prescription Administration Measures and the Regulations on the Administration of Narcotic Drugs and Category I Psychotropic Substances in Medical Institutions. Specifically, confirm that light red prescription forms are used and that the patient's identification number, the name of the proxy, and the proxy's identification number are clearly indicated. If a prescription includes <i>Pericarpium Papaveris</i> (poppy shell) without combination with other herbs, exceeds a 3-day dosage, surpasses the usual adult single dose of 6 g, or involves more than 7 consecutive days of administration, a clear alert should be issued and the physician advised to modify the prescription. If clinical necessity is confirmed, the prescription may be approved only after dual physician signatures. (2) Verify whether prescriptions involving toxic Chinese herbal medicines comply with the Prescription Administration Measures and the Measures for the Administration of Toxic Drugs for Medical Use. If the oral dosage exceeds the maximum specified amount, or if a single prescription exceeds the maximum two-day dosage, an explicit prompt should be issued recommending that the physician revise the prescription. If the physician declines to make modifications, the prescription should be rejected during the prescription audit.
	Standardized Guidelines for the Prescription Audit of Chinese Herbal Medicines	N-008	Verify whether the prescribed Chinese herbal medicines are documented on a separate prescription form and not combined with Western pharmaceuticals, proprietary Chinese medicines, or Chinese herbal formula granules on the same prescription sheet.
	Prescription Standards for Neonates and Infants	N-009	For neonates and young infants, prescriptions must specify the age in days and/or months, and indicate body weight when clinically necessary.
Appropriateness	Concept of Appropriateness in the Prescription Audit of Chinese Herbal Medicines	A-001	This pertains to key concepts including rational medication use, adverse drug reactions, and drug-induced diseases.
	Consistency Between Medication Use and TCM Diagnosis (Disease Name and Syndrome Pattern)	A-002	The alignment of prescribed Chinese herbal medicines with the TCM diagnosis (disease name and syndrome pattern) is classified into two types of mismatches: inappropriate indication and use without indication. Inappropriate indication occurs when the prescribed herbs align with the disease name but not the syndrome pattern. Use without indication refers to prescriptions that do not correspond to the disease, its complications, or the syndrome pattern.
Continued			

Audit Dimension	Prescription Audit Rules	Rule Identifier ^a	Rule Description
	Dosage Limits for Single Chinese Herbal Medicines	A-003	The dosage of individual Chinese herbal medicines should comply with the specifications outlined in the Pharmacopoeia of the People's Republic of China, the National Processing Standards for Chinese Herbal Medicines, and relevant provincial standards. In general, the dosages are adjusted according to age: neonates are typically prescribed one-sixth of the adult dose; infants receive one-third to one-half; young children are given two-thirds or the full adult dose; and school-age children generally receive the full adult dose. For elderly individuals over 60 years of age, the recommended dosage is one-half to three-quarters of the adult dose. When the prescribed dosage of a non-toxic or toxic Chinese herbal medicines exceeds three times the upper limit specified by these standards, the system should explicitly alert the physician and recommend modifying the prescription. If the clinical condition necessitates such a dosage, the prescription may only be approved after dual-signature confirmation by two physicians. If the dosage of individual toxic Chinese herbal medicines (i.e., a herb classified as toxic under TCM regulatory standards) exceeds the upper limit specified in the Pharmacopoeia of the People's Republic of China, the National Processing Standards for Chinese Herbal Medicines, or relevant provincial standards, or if the total amount in a single prescription exceeds the maximum two-day dosage, the system must generate a clear alert and recommend that the physician revise the prescription. If a higher dosage is clinically necessary, the prescription may be approved only after dual-signature confirmation by two physicians. However, if the dosage of a single legally toxic herbal piece—as defined by national regulations—exceeds the upper limit specified in the aforementioned standards, the system must issue a strict warning and require the physician to revise the prescription. If the physician refuses to modify the prescription, it must not be approved, even with dual-signature confirmation.
	Total Dosage and Ingredient Count Limits	A-004	The total dosage of Chinese herbal medicines in a single prescription should generally be controlled within 240 grams. As a general rule, if the total exceeds 300 grams (excluding paste prescriptions), an explicit alert should be issued, and the physician should be advised to revise the prescription. If a higher dosage is clinically necessary, the prescription may be approved only after dual-signature confirmation by two physicians. If local governmental authorities have issued specific regulations on the number of individual herbal ingredients permitted per prescription, the prescription audit should comply with those local requirements. In the absence of such regulations, the number of herbal components in a single prescription should generally be limited to 18 or fewer (excluding paste prescriptions).
	Rationality of Prescription Usage	A-005	The prescription audit shall verify the appropriateness of the dosage form, route of administration, dosing schedule, frequency, and treatment duration for Chinese herbal medicines. For herbal medicines not suitable for oral decoction use, a clear warning must be issued recommending prescription modification. If clinical necessity requires their use, the prescription may be approved after dual physician signatures.
	Appropriateness of Species Selection	A-006	The prescription audit shall verify the appropriateness of the selected botanical origins and processed forms of Chinese herbal medicines, with special attention to the rationale for herb selection in special populations including pediatric patients, elderly individuals with multiple chronic conditions, patients with hepatic or renal impairment, and lactating women.
	Contraindications for Use	A-007	Contraindications by Syndrome, Contraindicated Combinations, and Pregnancy Contraindications
	Identification of Duplicate Medications	A-008	The prescription audit shall verify whether the same Chinese herbal medicines is used multiple times in a single prescription, including cases involving: (1) duplicate use of the identical herb, (2) simultaneous use of different processed forms of the same herb, or (3) administration of different botanical origins classified under the same herbal name.
	Duplicate Medication and Herb Incompatibilities in Combination Therapy	A-009	The prescription audit should evaluate whether the prescribed Chinese herbal medicines exhibit therapeutic duplication or incompatibility with concurrently prescribed proprietary Chinese medicines or Western pharmaceuticals.

Table 1. Summary of the 18 prescription audit rules for Chinese herbal medicines. a C, N, and A represent the three audit dimensions: Fundamental Concepts, Normativity, and Appropriateness, respectively.

publicly available in structured digital form online. Accordingly, they represent non-open-source corpora, which helps mitigate the risk of training-test contamination and evaluation leakage due to potential overlap. To ensure thorough rule coverage, every question was explicitly mapped to a corresponding prescription audit rule.

In light of the need to ensure both thoroughness and meaningful differentiation in evaluation, we employed a two-pronged strategy during question development and collection:

1. **Breadth of Coverage:** We ensured that all key assessment points specified in the prescription audit rules were covered by corresponding questions. (1) For each audit rule involving a single assessment point, at least one question was developed to evaluate that specific point; (2) For rules comprising multiple distinct assessment points, separate questions were designed for each point.
2. **Depth of Assessment:** For audit rules associated with high clinical error rates, significant safety risks, or complex knowledge structures, we further designed clinical application questions. These questions simulated real-world scenarios. They were used to assess whether the model could apply its understanding of the rules in practice-oriented decision making.

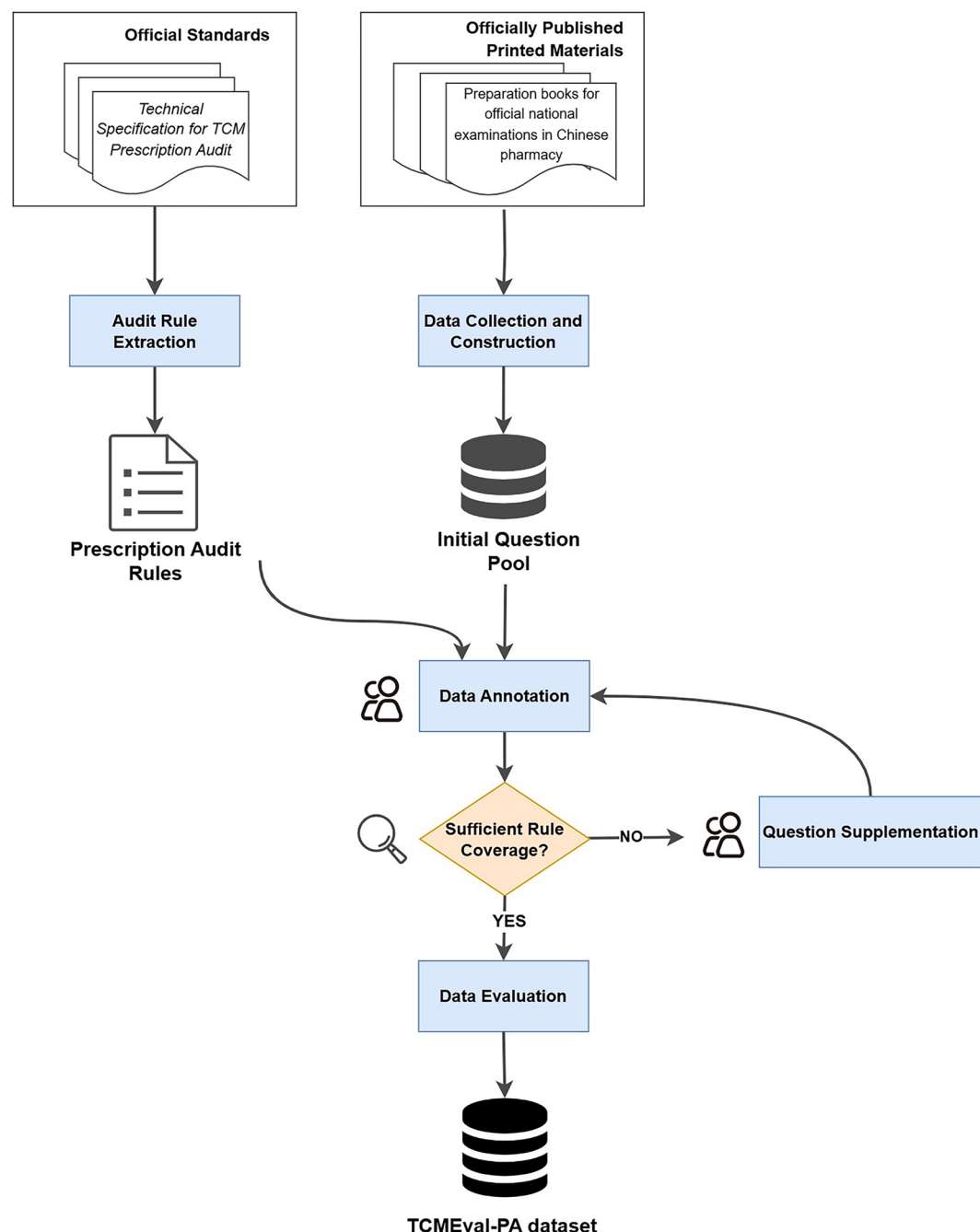


Fig. 1 Workflow for the construction of the TCMEval-PA dataset. Audit rules were first extracted from official standards. In parallel, question items were sourced from officially published printed preparation books for official national examinations in Chinese pharmacy to build the initial question pool. Each item was then annotated with its corresponding audit rule. Rule coverage was evaluated, and supplemental items were developed in cases of insufficient coverage. All questions subsequently underwent quality evaluation focusing on accuracy, clarity, and alignment with exclusion criteria. Only those meeting all evaluation standards were retained in the final dataset.

For example, consider the rule that governs the use of toxic Chinese herbal medicines, as outlined in official standards. “If the dosage of a single toxic herb exceeds the upper limit specified in the *Pharmacopoeia of the People’s Republic of China*, the *National Standard for Processing of Chinese Herbal Pieces*, or corresponding provincial standards, or if a single prescription exceeds the maximum two-day dosage, the prescription should be explicitly flagged and modification should be recommended. If use is deemed clinically necessary, the prescription may be approved after confirmation with dual physician signatures.” Building on this rule, we designed two levels of questions to evaluate the model’s capabilities from both comprehension and application perspectives. (1) The first was a rule comprehension question, aimed at assessing whether the model accurately understood

the triggering conditions for audit specified in the rule. This is exemplified by the rule comprehension item presented in Table 2; (2) The second was a knowledge application question. It focused on evaluating the model's understanding of commonly used toxic Chinese herbal medicines and its ability to integrate clinical knowledge with audit rules to make appropriate judgments.

Data annotation. To ensure systematic organization and comprehensive coverage of prescription audit rules, each question in the dataset was annotated with its corresponding rule identifier and classification dimension (either prescription normativity or prescription appropriateness). Each question was specifically mapped to its corresponding audit rule as listed in Table 1. As part of the annotation process, two licensed TCM physicians independently reviewed and labeled each question. After completing their independent annotations, the two practitioners conducted a cross-comparison to identify any discrepancies. Any disagreements were resolved through consultation with a licensed TCM pharmacist, who adjudicated and finalized the annotation. This rule-based annotation framework enables fine-grained analysis of model performance across the two dimensions and provides a solid foundation for subsequent evaluation of model capabilities.

Question supplementation for rule coverage. Throughout the process of question collection and annotation, we continuously monitored the coverage of prescription audit rules across the dataset. When a given rule had no associated question, or when the existing items failed to capture the rule's structural complexity or clinical significance, it was flagged as a case of insufficient rule coverage. In such cases, supplemental questions were developed by the expert team according to the following principles. First, priority was given to designing questions that addressed common issues in clinical prescriptions, thereby enhancing both representativeness and practical relevance. Second, the structure and language of the newly created questions were kept consistent with those of the original question pool to ensure stylistic uniformity.

To align the dataset with real-world clinical needs, we exercised strict control over content scope throughout the question collection and design process. This approach ensured that the dataset accurately reflects the practical requirements and task complexity of clinical prescription audit. As a result, the dataset offers a high-quality and structured foundation for evaluating the capabilities of LLMs in this specialized task scenario.

Data evaluation. After the annotation process was finalized, all 328 questions included in the study were subjected to a rigorous quality evaluation. The assessment focused on three core aspects: accuracy, clarity, and exclusion criteria. First, for accuracy, all correct answers were individually validated by licensed TCM physicians to ensure that each was authoritative and unambiguous. Second, clarity was evaluated by reviewing the wording of each question stem and its corresponding options. Questions containing ambiguity, logical flaws, or redundant distractors were eliminated. Third, questions were excluded if they contained factual errors, lacked a clearly correct answer, had vague assessment points, or showed unclear mappings to the corresponding audit rules.

Only those questions that met all of the following criteria—clear semantics, a single correct answer, and complete annotation with rule identifiers and classification dimensions—were included in the final version of the TCMEval-PA dataset. The distribution of the 328 finalized questions across the 18 audit rules and their associated classification dimensions is summarized in Table 3. These quality control measures ensure the scientific rigor, reliability, and reproducibility of the dataset for downstream model evaluation tasks.

Data Records

The TCMEval-PA dataset is available at Figshare³¹ under the CC BY 4.0 license and is provided in .xlsx format. All questions were independently developed and annotated by licensed TCM physicians, following official national standards and technical specifications for the prescription audit of Chinese herbal medicines. Each item is explicitly linked to a specific audit rule and includes an explanatory note to ensure the authority and traceability of the data.

The TCMEval-PA dataset consists of 328 choice questions, including 297 single-choice and 31 multiple-choice items. The questions are categorized as follows:

1. Normativity: 86 questions corresponding to 9 specific rules;
2. Appropriateness: 236 questions corresponding to 9 specific rules;
3. Fundamental Concepts of prescription auditing: 6 questions.

The dataset contains the following fields, with full definitions provided in Table 4: ID, Question, Candidate Answers, Answer, Explanation, Category, Rule ID, and Rule Summary.

Technical Validation

Experimental design. To evaluate the practical utility and discriminative power of the TCMEval-PA dataset in assessing the prescription auditing capabilities of LLMs, we selected several representative models for evaluation. These included general-purpose models (GPT-4o³², DeepSeek R1³³, and Qwen-Plus 2025-04-28³⁴) and TCM-oriented models (HuatuoGPT³⁵, ShenNong-TCM-LLM³⁶).

All questions were presented to the models in multiple-choice format, with prompts used to guide responses. The prompt design was adapted to the specific type of question:

- 1) Single-choice questions: "The following is a single-choice question on the prescription audit of Chinese herbal medicines. Please choose the correct option according to the rules. Output only the option letter."
- 2) Multiple-choice questions: "The following is a multiple-choice question on the prescription audit of Chinese herbal medicines. Please choose all correct options according to the rules. Output only the option letters."

Design Dimension	Question Design Description	Question	Correct Answer
Rule Comprehension	Design questions that comprehensively address all key knowledge points specified in the audit rule to achieve breadth of coverage.	Which of the following most completely and accurately describes the triggering conditions for audit regarding toxic Chinese herbal medicines? A. Only when the single-herb dosage exceeds the upper limit specified in the ChP B. Only when the total prescribed amount exceeds the maximum two-day dosage C. When the single-herb dosage exceeds the ChP limit or the prescription exceeds the maximum three-day dosage D. When the single-herb dosage exceeds the upper limit or the total prescribed amount exceeds the maximum two-day dosage E. When the single-herb dosage exceeds the daily recommended amount stated in the package insert	D
Knowledge Application	Focusing on clinical application scenarios associated with the rule to ensure a thorough evaluation of the model's depth of understanding and its ability to make clinically sound judgments.	What is the recommended dosage range for decoction of Toosendan Fructus (Chuan Lian Zi) when taken orally? A. 1–3 g B. 3–6 g C. 5–10 g D. 6–12 g E. 10–15 g	A

Table 2. Sample questions based on the rule for toxic Chinese herbal medicines.

Audit Dimension	Rule Identifier	Count
Fundamental Concepts	C-001	6
Normativity	N-001	11
	N-002	10
	N-003	36
	N-004	1
	N-005	5
	N-006	6
	N-007	14
	N-008	1
	N-009	2
Appropriateness	A-001	35
	A-002	4
	A-003	87
	A-004	9
	A-005	20
	A-006	9
	A-007	64
	A-008	4
	A-009	4

Table 3. Distribution of questions in TCMEval-PA by audit dimension and rule identifier.

Model responses were collected via API calls. A strict scoring criterion was applied: a response was marked correct only if the set of options produced by the model exactly matched the reference answer; otherwise, it was marked incorrect.

Result analysis. The evaluation results are summarized in Table 5 below. Overall, general-purpose LLMs outperformed TCM-oriented models, particularly in their ability to understand and apply rules related to prescription normativity and appropriateness.

Among the general-purpose models, Qwen-Plus (2025-04-28) achieved the highest overall performance, with an accuracy of 86.05% (74/86) in the prescription normativity dimension. DeepSeek R1 performed consistently well, maintaining accuracy above 74% in both dimensions. The comparatively lower accuracy of GPT-4o in the prescription appropriateness dimension suggests limitations in its ability to handle contraindicated combinations, dosage control, species selection, and administration methods. By contrast, TCM-oriented models performed significantly worse. Neither HuatuoGPT2-7B nor ShenNong-TCM-LLM achieved the expected results, showing especially low accuracy in the prescription normativity. Although these models were trained on large-scale domain-specific corpora, their advantages have yet to materialize in safety-critical tasks. Results from the fundamental concepts are reported for reference only due to the limited number of items.

Taken together, the findings indicate that although TCM-oriented LLMs have been fine-tuned on specialized domain corpora, their performance on tasks involving rational medication use and prescription safety remains inferior to that of general-purpose models. These results reinforce the core motivation of this study: to establish a systematic, task-driven evaluation framework capable of assessing model performance in high-risk, rule-intensive medical tasks and identifying critical gaps in medical knowledge accuracy and reasoning capacity.

Field Name	Description
ID	Unique identifier for each question
Question	The full question text, including context and content
Candidate Answers	List of candidate options
Answer	Correct answer(s); may be single-choice or multiple-choice
Explanation	Justification and reasoning for the correct answer
Category	Question type: Normativity, Appropriateness, or Fundamental Concepts
Rule ID	Unique identifier of the corresponding prescription audit rule (e.g., “A-001”, “N-005”)
Rule Summary	Brief title summarizing the specific prescription audit rule tested in the question. These titles correspond to the “Prescription Audit Rule” column in Table 1.

Table 4. Description of data fields in TCMEval-PA.

LLMs	Normativity n = 86, n (%)	Appropriateness n = 236, n (%)	Fundamental Concepts n = 6, n (%)
GPT-4o	61 (70.93%)	148 (62.71%)	5 (83.33%)
DeepSeek R1	67 (77.91%)	176 (74.58%)	5 (83.33%)
Qwen-Plus (2025-04-28)	74 (86.05%)	182 (77.12%)	6 (100%)
HuatuoGPT2-7B	30 (34.88%)	86 (36.44%)	2 (33.33%)
ShenNong-TCM-LLM	7 (8.14%)	41 (17.37%)	1 (16.67%)

Table 5. Accuracy of LLMs across different prescription audit dimensions.

Usage Notes

The evaluation conducted in this study revealed that current TCM-oriented LLMs perform suboptimally in tasks related to prescription safety. This highlights the urgent need for a standardized dataset grounded in prescription audit rules to support both the evaluation and improvement of model performance. The core value of TCMEval-PA lies in its use of structured, rule-annotated choice questions to enable comprehensive, rapid, and transparent assessment of LLMs’ capabilities in the prescription audit of Chinese herbal medicines. The dataset provides systematic coverage of all 18 audit rules and adopts a two-tier item design—rule comprehension and scenario-based application—targeting high-risk and complex aspects of prescription safety. In addition to enabling horizontal comparisons across different models, TCMEval-PA also provides an evidence-based foundation for improving the safety-related capacities of TCM-oriented LLMs.

Existing safety-related benchmarks either generalize across the entire medical domain—failing to capture the unique regulatory framework of Chinese medicine—or focus on TCM-specific tasks while lacking clearly defined task boundaries and authoritative rule mappings^{23,37–39}. TCMEval-PA is the first benchmark to center specifically on the prescription audit rules of Chinese herbal medicines. It establishes a safety evaluation framework that encompasses both structural requirements and medication use constraints, offering a unified and quantifiable reference standard for enhancing the prescription safety capabilities of LLMs. This QA dataset offers three key advantages:

- 1. **Comprehensiveness:** Every prescription audit rule is represented by at least one corresponding item, ensuring complete evaluation coverage with no gaps.
- 2. **Clarity:** Each question is paired with a definitive correct answer and an explicit rule mapping, allowing for transparent verification and enabling consistent comparisons across different models and versions.
- 3. **Targeted Insight:** By quantifying model performance across distinct dimensions, the dataset facilitates the identification of weaknesses and supports targeted model optimization.

It is important to note that TCMEval-PA does not aim to replicate the full clinical prescription audit workflow. Rather, it serves as an intermediate benchmark focused on prescription safety, providing a reproducible and extensible starting point for model evaluation. Future work will build upon TCMEval-PA to support tasks that more closely resemble real-world clinical scenarios, such as open-ended audits based on actual prescription texts and the joint identification of multiple safety violations. These efforts will contribute to the progressive construction of a multi-level evaluation framework for the prescription audit of Chinese herbal medicines.

Evaluation procedure and field usage. Users can independently assess model performance based on the two core dimensions of prescription audit: normativity and appropriateness. We recommend using the Category field to facilitate dimension-specific analysis.

The dataset is released in .xlsx format, with each question including the following fields: 'ID': Unique identifier for each question; 'Question': Full question text; 'Candidate Answers': List of candidate options; 'Answer': Correct answer(s); 'Explanation': Reasoning for the correct answer; 'Category': Question type; 'Rule ID': Identifier of the corresponding audit rule; 'Rule Summary': Brief title summarizing the audit rule tested in the question. Evaluation adopts a strict scoring criterion: a model’s response is marked as correct only if the selected option(s) exactly match the reference answer; otherwise, it is considered incorrect.

Data availability

The dataset is openly available on Figshare³¹ at <https://doi.org/10.6084/m9.figshare.29651261.v3> (version 3). Data are released under the CC BY 4.0 license with no access restrictions.

Code availability

The Python evaluation scripts are available in the same Figshare record as the dataset, under the **scripts/** folder³¹. This Figshare record is the version of record for reproducibility. A synchronized copy is available on GitHub (<https://github.com/zhuyan166/TCMEval/tree/main/evaluation/TCMEval-PA>).

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Competing interests

The authors declare no competing interests.

Additional information

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